WASP-12b

R-0162

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_250208` · created 2026-07-16T15:56:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460714.7865 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.045
Transit depth (Rp/Rs)^2	2.39 +/- 1.39 [%]
Orbital Inclination (inc)	85.41 +/- 2.86
Airmass coefficient 1 (a1)	22.85 +/- 20.1
Airmass coefficient 2 (a2)	-2.95 +/- 1.71
Scatter in the residuals of the lightcurve fit is	99.55 %
Best Comparison Star	#1 - [191, 183]
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.0764 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.045 vs 0.1178 (deviation 0.83σ)
Duration fit vs published	0.0764 d vs 0.125 d (deviation 13.5σ)
Mid-time O–C	-1.5 min
Depth SNR	3.4
Residual scatter	99.55 %

Light curve

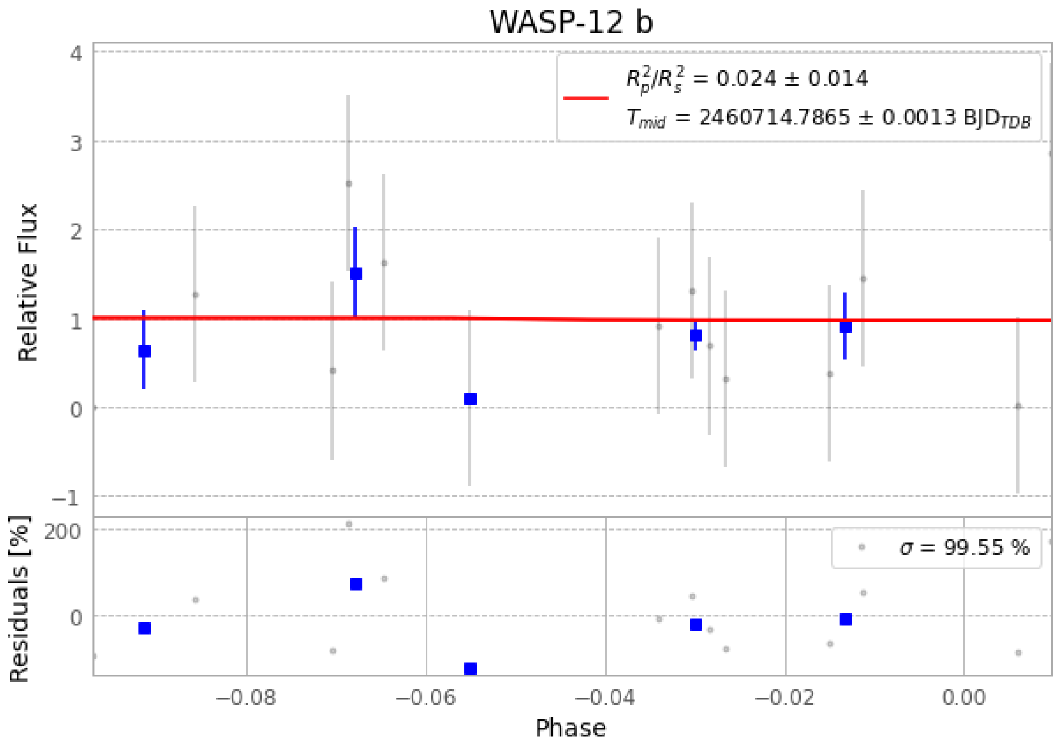

Figure 1 — FinalLightCurve_WASP-12 b_2025-02-07.png (SHA-256 461f7758ff24b94f…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [247, 201], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cec43932a35d06df…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1634555`

Data provenance

57 FITS frames (37 MB), `WASP-12250208040918.FITS` → `WASP-12250208070015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250208.log (e29afd50b453…), FinalLightCurve_WASP-12 b_2025-02-07.png (461f7758ff24…), FinalLightCurve_WASP-12 b_2025-02-07.csv (152946b1184c…)

Compute resources

Wall time 210.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-16 15:56Z.